

CARLOS IVAN SANDOVAL BARRON

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EDUCATION

Colorado School of Mines

Expected 05/2027

B.S. Computer Science — Cybersecurity Certificate

Golden, CO

- GPA: 3.98 | Dean's List
- Relevant Coursework: Foundations of Programming, Systems Programming, Data Structures and Algorithms, Software Engineering, Database Management, Programming Languages, Algorithms, Introduction to Cryptography, Advanced Software Engineering, Information Security & Privacy

EXPERIENCE

Security Engineer Intern

05/2026 – 07/2026

Spectrum

Denver, CO — Hybrid

- Built two internal tools spanning security and finance: a Python backend that remotely audits over 300 legacy devices for security-agent compliance, and a budget-forecasting dashboard that replaced a 3-5 business day manual data-consolidation process with one completed in minutes for company leadership.
- Applied parallelization to the compliance-checking backend, cutting per-device testing runtime from over an hour to minutes, and used MongoDB and SSH to track and test device state at scale.
- Adapted the forecasting dashboard's rules to handle real-world irregularities like employee turnover, PTO, and overtime after mock data proved too clean; the sponsoring business team requested continued work with real data after seeing the results.

Data Structures and Algorithms TA

08/2025 – Current

Colorado School of Mines

Golden, CO

- Assist students in understanding concepts they were not able to comprehend during lecture.
- Grade analysis projects from students, provide feedback, and explain areas of improvement.

PROJECTS

Agent Checker | Spectrum Internship — Python, MongoDB, SSH, HTML/CSS/JS

Summer 2026

- Built a tool to remotely connect to and verify compliance for over 300 legacy devices against required security agents.
- Used MongoDB and SSH to track device state, applying parallelization to cut testing runtime from over an hour to minutes.
- Built an HTML/CSS/JS dashboard pulling from the databases to display compliance results and metrics.
- Used Claude Code as part of the development workflow.

Budget Forecasting Dashboard | Spectrum Internship — Python, AWS Lambda, AWS S3

Summer 2026

- Forecasted company spend across resources — employees, AWS costs — using Python scripts on AWS, keeping operating and capital expenses separate.
- Built a dashboard with both a high-level and a detailed view for different levels of management, cutting what previously took 3-5 business days of manual data consolidation down to minutes.
- Delivered in a competition format against other intern teams; the sponsoring business team liked the results enough to request continued work with real data.

De-Bug | Colorado School of Mines Hackathon — 1st Place, Game Development — C++, raylib, CMake

Spring 2026

- Built a 3D platforming/puzzle game from scratch in C++ using raylib on a 4-person team, winning 1st place in the game development track.
- Designed a custom, data-driven level loader that reads structured text files to generate platforms, objects, and interactions, instead of hardcoding levels.
- Implemented an in-game terminal where players solve programming/logic puzzles to progress through levels.

Autonomous Trading Research Agent | Personal Project — Python, Claude Code, Robinhood API

2026

- Built an autonomous agent that researches market, economic, and political conditions and places real equity trades on a live brokerage account on a daily schedule.
- Designed risk-management rules around real regulatory constraints to keep the agent's trading behavior compliant and safe.
- Runs a weekly retrospective on past trades to catch and fix its own reasoning errors, refining its rules over time.
- Redesigned its higher-risk strategy after a rough first month, now requiring stricter evidence before acting.

ORGANIZATIONS

Society of Hispanic Professional Engineers

Member since 08/2024

Outreach Chair

08/2026 – Current

- Co-lead the outreach program and design curriculum for high school students to build interest in the STEM field.
- Tutor and mentor students in the junior chapter at two local high schools.
- Coordinate events and activities for younger students to introduce them to STEM.
- Contribute to organizational leadership through weekly Executive Board meetings.

SKILLS & ABILITIES

Technical: Java, C++, Python, C, SQL, MongoDB, AWS Lambda & S3, SSH, Linux Command Line, JUnit, Claude Code, Microsoft Office, LaTeX, GitHub

Professional: Time Management, Customer Service, Adaptability, Innovative, Performance Tuning, Leadership, Team Worker, Fluent Spanish, Technical Writing